Ex. No. 3	ADD INTERACTIVITY TO APP – INPUT AND FORMS
Date of Exercise	24/08/2026
GitHub Host Link	

Aim

To create an interactive Flutter student registration form with input fields, form validation, and a submit button.

Description

This experiment demonstrates how to create an interactive form in Flutter using `Form`, `TextFormField`, `GlobalKey<FormState>`, and `ElevatedButton`. The form accepts student name, email, and password. Validation messages are displayed when required fields are left empty.

Procedure:

1. Create a new Flutter project using the command:

   flutter create student_form

2. Open the project folder:

   cd student_form
   code.
3. Open the lib/main.dart file.
4. Create a GlobalKey<FormState> to manage form validation.
5. Create a Form widget and assign the form key.
6. Add a TextFormField for entering the student name.
7. Add an email input field using TextInputType.emailAddress.
8. Add a password field using obscureText: true.
9. Add an ElevatedButton for submitting the form.
10. Use the validate() method to check the entered information.
11. Display a success message using SnackBar when the form is valid.
12. Run the application and test the validation by clicking Submit without entering any information.

Program

```
import 'package:flutter/material.dart';

void main() {
  runApp(const StudentFormApp());
}

class StudentFormApp extends StatelessWidget {
  const StudentFormApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
    debugShowCheckedModeBanner: false,
    home: const StudentForm(),
    );
  }
}

class StudentForm extends StatefulWidget {
  const StudentForm({super.key});

  @override
  State<StudentForm> createState() => _StudentFormState();
}

class _StudentFormState extends State<StudentForm> {
  final _formKey = GlobalKey<FormState>();

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Student Registration'),
      ),
      body: Padding(
        padding: const EdgeInsets.all(20),
        child: Form(
          key: _formKey,
          child:  Column(
          children: [
              TextFormField(
                decoration: const InputDecoration(
                  labelText: 'Student Name',
                  border: OutlineInputBorder(),
                ),
```

```
  validator: (value) {
    if (value == null || value.isEmpty) {
      return 'Enter student name';
    }
    return null;
  },
),

const SizedBox(height: 15),

TextFormField(
  decoration: const InputDecoration(
    labelText: 'Email',
    border: OutlineInputBorder(),
  ),
  keyboardType: TextInputType.emailAddress,
  validator: (value) {
    if (value == null || value.isEmpty) {
      return 'Enter email';
    }
    return null;
  },
),

const SizedBox(height: 15),

TextFormField(
  obscureText: true,
  decoration: const InputDecoration(
    labelText: 'Password',
    border: OutlineInputBorder(),
  ),
  validator: (value) {
    if (value == null || value.isEmpty) {
      return 'Enter password';
    }
    return null;
  },
),
```

```
                const SizedBox(height: 20),

                ElevatedButton(
                  onPressed: () {
                    if (_formKey.currentState!.validate()) {
                      ScaffoldMessenger.of(context).showSnackBar(
                      const SnackBar(
                          content: Text('Form submitted successfully'),
                        ),
                      );
                    }
                  },
                  child: const Text('Submit'),
                ),
              ],
            ),
          ),
        ),
      );
    }
  }
```

Output

localhost:3000
Student Registration
Student Name
Student Name
Department
Submit
Elements
Console
Network
Performance
Memory
Application
Styles
Computed
Layout
Event Listeners
Properties
Accessibility
Console
style.css:3
style.css:11

Result

Thus, an interactive Student Registration Form was successfully created using Flutter. The application contains Name, Email, Password, Validation, and Submit Button, and displays appropriate validation messages for empty fields.